

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 1: Image Processing Pipeline

PART 1: CORRECT ORDER OF IMAGE PROCESSING PIPELINE

1. **Image Acquisition**
2. **Pre-processing**
3. **Segmentation**
4. **Feature Extraction**
5. **Classification**

PART 2: PURPOSE OF PRE-PROCESSING

The primary purpose of pre-processing is to enhance raw image quality by suppressing unwanted noise, removing distortions, and improving contrast and visual detail. It standardizes the data (e.g., via normalization, resizing, or grayscale conversion) so that subsequent high-level pipeline stages—such as segmentation, feature extraction, and classification—can operate more accurately and efficiently.

KEY FUNCTIONS OF PRE-PROCESSING

- **Noise Reduction & Filtering:** Attenuates random sensor noise using spatial or frequency domain techniques (e.g., median or Gaussian filters).
- **Image Contrast Enhancement:** Adjusts dynamic range and luminance levels (e.g., histogram equalization) to make key visual features distinct.
- **Data Normalization & Formatting:** Resizes images to standard target dimensions and normalizes pixel intensity distributions.
- **Geometric & Radiometric Corrections:** Corrects optical lens distortion, perspective tilt, and non-uniform lighting conditions.